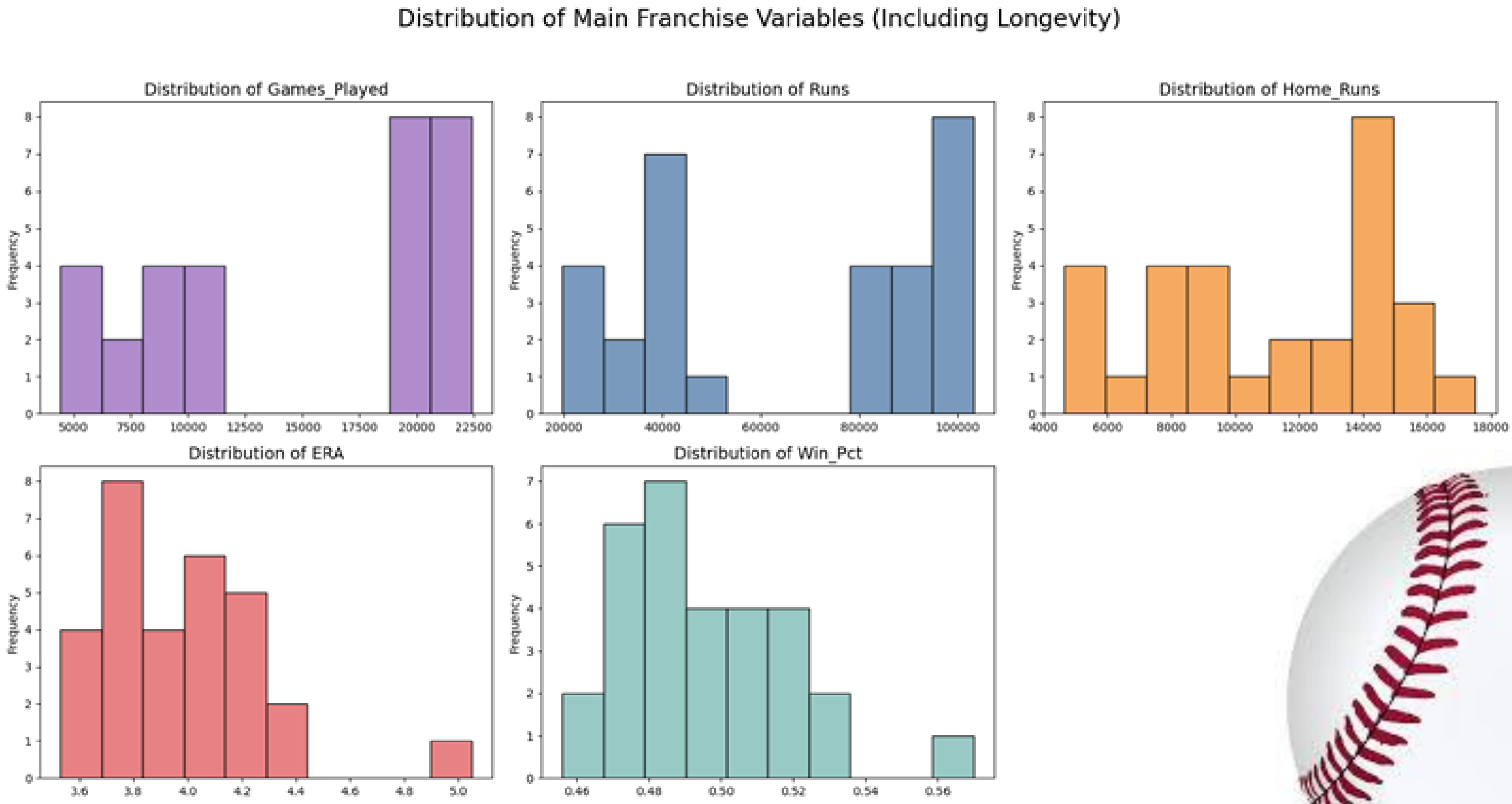
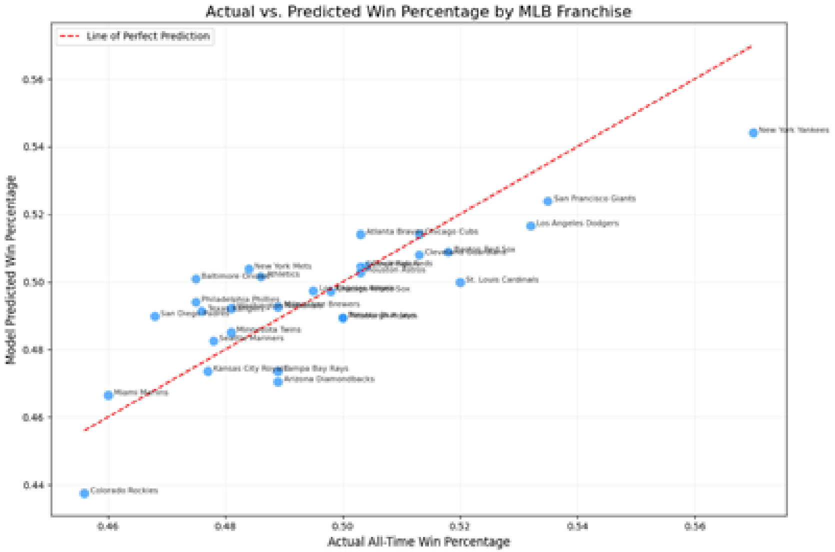
Is hitting or pitching more important for becoming a top-tier MLB franchise?



Cross-sectional data of all 30 MLB franchises from the Lahman Database

Statistical Model: ***Control Variable-Games Played***

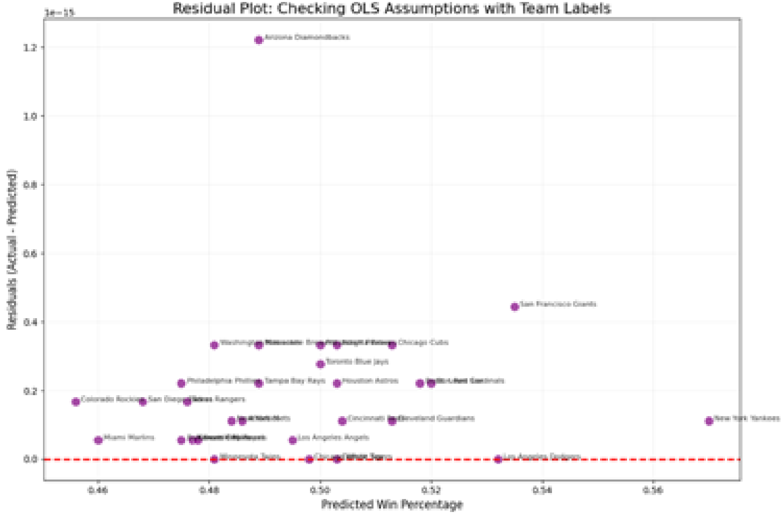
Win_Pct = beta_0 + beta_1(Games) + beta_2(Home Runs) + beta_3(ERA)



Pitching (ERA): For every 1-unit increase in career ERA, win percentage drops by 5.07 percentage points (beta 3 = -0.0507, p = 0.001).

Hitting (HR): For every additional home run hit, win percentage increases by 0.00076 percentage points (beta 2 = 0.0000076, p = 0.002).

The plot shows residuals randomly distributed around zero, confirming that all OLS assumptions are satisfied and the results are statistically reliable.



While both hitting power and pitching efficiency are significant drivers of success, the magnitude of the ERA coefficient proves that elite pitching is the most critical factor in achieving a higher historical win percentage.